# Game Manual

## Your first patrol aboard a Vanguard-class frigate

Cold fusion arrived around 2050. Five decades later, Electromagnetic Pulse Drives connect the Solar System. Your watch begins at Earth on 1 January 2100.



VANGUARD-CLASS FRIGATE / SIDE PROFILE

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# First watch

Start at Earth, choose a destination, and take the ship out on its first EPD leg.

## Launch and plot a course

01 **Launch a patrol.** Choose **New Mission**, name your Vanguard-class frigate, and select **Launch mission**. Use **Continue** for an existing browser patrol. A new mission replaces the browser autosave.

02 **Choose a destination.** Open **Flight Control / System Map**. Select a planet, settlement, mining station, or point in open space. Choose **Add to course**. You can queue up to 12 waypoints.

03 **Engage EPD.** Check the route and estimated travel time, then select **Engage EPD**. If the route crosses Sol's exclusion zone, clear or edit the plan and add a detour around Sol before the crossing leg.

> TRANSIT CONTROLS
>
> **1×, 10×, 100×** change the EPD simulation rate. **Pause** holds the trip. **Abort transit** returns you to local flight in deep space with the remaining waypoints queued.

## Bridge stations

| | |
|---|---|
| **Bridge Command** | Review hull readiness, alerts and shuttle status. Open a station from the bridge deck. |
| **Flight Control** | Plan strategic routes in **System Map** and handle nearby flight in **Local Flight**. |
| **Tactical / Security** | Read sensors, lock contacts, maneuver, fire weapons and launch shuttles. |
| **Comms** | Read dispatches, hail ships or stations, request orders and recall shuttles. |
| **Engineering / Support** | Contain emergencies, manage repair teams and request docking or service. |

# Comms and assignments

Sol Command and stations issue random assignments while you are on patrol. One is open at a time.

## From message to mission site

01 **Open the dispatch.** Choose **Open Comms** on the incoming popup. **Dismiss** closes the popup and leaves the message unread. Use the top-bar **Comms** button to return to unread messages.

02 **Plot to the red X.** Reading the message reveals a red **X** on the System Map. Select it or use **Show on System Map** in the message. Choose **Add to course**, then **Engage EPD**.

03 **Check the situation on arrival.** The ship drops out of EPD at the assignment and keeps later waypoints queued. Local time starts paused. Read the encounter report in **Tactical** before running the simulation.

## Assignment objectives

| | |
|---|---|
| **Station under attack** | Repel the raiders and keep the station's evacuation transport alive. |
| **Cargo ship under attack** | Stop the raiders before they destroy the freighter. Incoming shots can remain in flight after an attacker stands down. |
| **Sector patrol orders** | Sweep with radar operating and investigate unusual contacts.<br>In a quiet sector, watch Comms for further orders. |
| **Missing freighter** | Find the disabled ship and approach within 10 km with radar operating to confirm its position. |

## Reply, decline, or request orders

Select a channel and a **Request / response**, then choose **Transmit message**. Before arrival you can **Decline assignment**. Leaving before clearance or mission completion can result in a failed assignment. Comms records the result.

Once cleared, keep sweeping or return to Earth. If you stay, watch **Comms** for a new assignment or return orders. Use **Request orders** on the Sol Command channel.

# Local flight and sensors

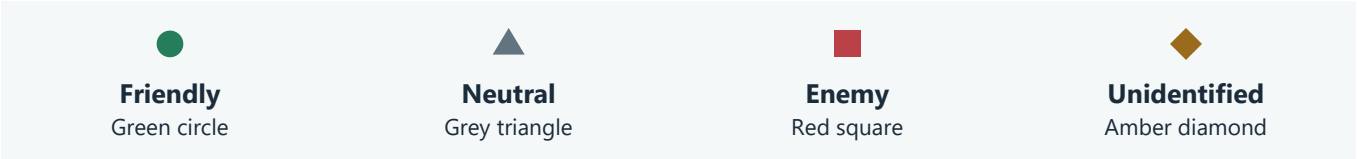
Local Flight handles movement within a sector. Tactical uses the same movement and contacts.

## Fly and stop

In **Flight Control / Local Flight**, click the sector chart to add up to eight stops and select **Fly route**. If the autopilot brakes for the training debris field, plot a waypoint around it. Use **Stop ship** to brake. **Cancel maneuver / coast** keeps your current velocity.

Use the top bar to **Run** or **Pause** local time. The rates are **0.5×, 0.75×, 1×, 10×, 60×**. A new flight or firing order can start local time. Use 1× to assess a fight or a close approach.

## Read the radar




Tactical radar offers 10 to 200 km display ranges. IFF can identify a ship within 100 km, with a stable chance of success per track. Hover or focus a track for 3D distance, elevation, and closure. Positive **Vc** means closing. A hollow last-known marker is a stale track.

## Lock and maneuver

01 **Lock a contact.** Select a live track on Radar or HUD and confirm a target priority. **1** is highest. Ctrl-click adds or releases a lock while keeping the others. Use **Targets / Tracking** to review priorities.

02 **Set a maneuver.** Choose **Intercept** for a 5 km standoff, **Match velocity** to move with the ship, or **Maintain range**, **Open distance**, or **Break away** to control separation.

**V** opens the HUD. The four directions cover forward, port, starboard and rear. Locks are shared across the HUD and the Tactical displays.

# Tactical combat

Identify the threat and check range, sensor quality and ammunition before committing to an engagement.

## Use automatic or manual fire control

01 **Assign targets.** Open **Tactical / Security**, lock identified hostile ships, and set their priorities. The **Weapons** display shows mounts, target assignments and aim systems.

02 **Order the engagement. Engage ship(s)** follows target priority. Guns suppress enemy CIWS first. Then guns and MP missiles can suppress an active jammer before the ship attacks offensive weapons.

03 **Take manual control if needed.** Check the mounts and choose a target and aim system. **Fire** issues one firing action or a finite cannon burst. **Start firing** repeats the order for each assigned mount.

> BREAK CONTACT
>
> **Cease Fire** ends your firing orders. Shots already in flight continue. Choose **Break away** to increase separation, or **Pause** to review the situation.

## Frigate weapon ranges

| | |
|---|---|
| **120 mm autocannon** | **30 km.** Use for close-range fire and suppression. |
| **30 mm CIWS** | **20 km.** Automatic point defense intercepts incoming missiles when enabled. |
| **MP missiles** | **80 km.** Target exposed subsystems such as engines, weapons or jammer. |
| **AS missiles** | **200 km.** Deliver heavy strikes against hull or ship systems. |

Weapon ranges are nominal. Internal systems are protected until the hull reaches zero. Jamming can break precision tracking. Use **Radar burn-through** in **Defense / Systems** for a short tracking window. Right-click a friendly SDF ship for fire support commands.

# Boarding and ship service

Use shuttles for nearby ship operations and Engineering for your own frigate's damage and stores.

## Launch and recover shuttles

01 **Approach within 10 km.** Lock the target on Radar or HUD. Open **Defense / Systems** and choose a shuttle order.

02 **Choose the job. Capture ship** sends marines to seize a ship. **Explore / search** reports its condition. **Assist crew** contains fire and seals pressure loss. Each launch sends 10 marines.

03 **Recover both shuttles before EPD.** Teams return after completing the job. Use **Comms / Shuttle control** to recall a team. If capture meets resistance, weaken the target, obtain surrender, reinforce, or recall.

## Repair, dock and replenish

| | |
|---|---|
| **Onboard repairs** | In **Engineering**, isolate emergencies first. Automatic repairs normally queue eligible damage. Field repairs consume spares and stop at 60% hull and 75% equipment. |
| **Dock and request service** | At a service site, hold within 5 km of the arrival point below 10 KPH with the area clear of immediate threats. Choose **Request docking**, then **Request station repairs and replenishment**. |
| **Full service** | Moon stations, Heliopause station and Mars colony can replace destroyed equipment, fully repair the ship, and replenish ammunition and spare parts. |
| **Mining stations** | Mining stations provide limited field repairs, spare parts and gun ammunition. Use a full-service site for missile stocks and destroyed systems. |
| **Repair tender** | Use **Request resupply / repair ship** in Engineering or through HQ or station Comms. Run local time and hold below 10 KPH for a safe rendezvous. Replenishment follows repairs. |

# Saves and quick reference

Keep a portable patrol backup and check the controls when a command is unavailable.

## Keep your patrol

The browser autosaves your patrol. Choose **Saves / Save game (.json)** to download a portable copy. To restore it, open **Load Saved Game** or **Saves**, choose the JSON file, and select **Load selected save**.

Loading resumes flight paused. Saved progress includes position, courses, damage, ammunition, locks, orders and active assignments. Returning to **Menu** also pauses the ship. Export a backup before starting over.

## Keyboard shortcuts

| | |
|---|---|
| **B / F / T** | Bridge Command / Flight Control / Tactical |
| **C / E / S** | Comms / Engineering / Science |
| **V / Esc** | Open the HUD / close the active HUD or popup |
| **Enter / Space** | Activate the focused control or acquire a focused track. |
| **Ctrl + Enter / Space** | Toggle a focused lock while retaining other locks. |

## Common blockers

| | |
|---|---|
| **No red X** | Read the assignment in Comms. Open **Show on System Map** to center the overview and select the fix. |
| **EPD unavailable** | Check for a safe plotted course, an operational EPD, deployed shuttles, docking or a connected repair tender. |
| **Weapons hold fire** | Check range, ammunition, mount condition, power and tracking. Automatic fire can hold missiles while enemy CIWS survives. |
| **No local progress** | Check the top-bar clock. Run local time for maneuvers, combat, recovery or repairs to advance. |